Markdown and webpages with knitr

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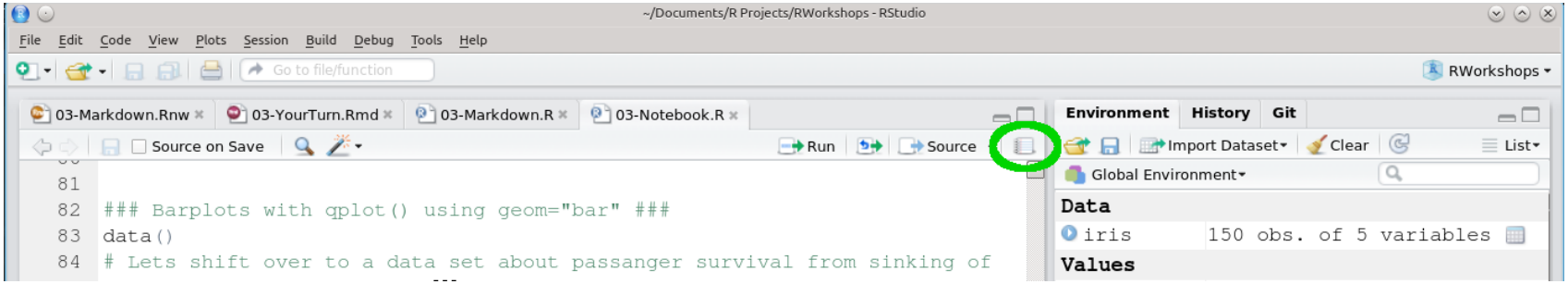
Knitr can also be used to create webpages

- Compile “notebooks" displaying your code and graphics in a single webpage
- Use knitr to create multi-purpose webpages
- R presentations with knitr and RStudio
- `knitrBootstrap` for more advanced HTML/Markdown options

R Code Notebooks

Rstudio will compile an R script into an HTML notebook that includes all generated pictures, output, and code.

This may make it easier for you to remember what a specific code file does and easily present your analyses.



What is markdown?

Markdown is a text-based format that can be easily converted into HTML.

```
Header 1
--------------------------------

*italics* and **bold** text

Lists are easy too:
- Item 1
- Item 2
- Item 3
```

R markdown

R markdown is very similar to the LaTeX code chunks we used earlier.
To create a code chunk (in any language), use

```
This is a code chunk
It will have fixed-width font
```

To create an R code chunk,

```{r chunkname, fig.width=7, fig.height=6}
x <- rnorm(100)
library(ggplot2)
qplot(x, geom="histogram")
```

Inline R code in Markdown

To make inline R code in markdown, use `` `r rnorm(20)` ``

This is analogous to `\Sexpr{}` in LaTeX documents

Protip: Click on the "?" button in the toolbar on a .Rmd document in Rstudio to get a Markdown cheat-sheet!

Equations in R Markdown

R Markdown also includes MathJax equations, which allow LaTeX equation syntax in web pages.
Single-line equations:

```
$x=y=z$
```

Multi-line equations:

```
$$
a x^2 + bx + c = 0\\
... = 0\\
x = \frac{1}{2a}\left(-b \pm \sqrt{b^2 - 4ac}\right)
$$
```

Some rules apply

Your Turn

- In RStudio, create a new R Markdown File (File $->$ New File $->$ R Markdown)
- Change the title of the document, add a couple of section headers, try out bold and italic text, make a table, and add an R code chunk
- In your R code chunk, save 100 randomly generated x values and plot a histogram
- Outside your code chunk, calculate the mean of the values you generated and reference that using an inline code chunk
- Compile your document

Tables and Markdown

`knitr` can be used to make HTML tables with the `kable` function:

```
library(knitr)
kable(dataframe)
```

`stargazer` will also create HTML tables for regression output:

```
stargazer(lm(y~x, data=dataframe), type="html")
```

Your Turn

- Print the first 6 lines of the iris data to a HTML table in a markdown document
  Don't forget `results='asis'` in the code chunk definition!
- Fit a linear regression using Petal Length to describe Sepal Length
- Output the results of the linear regression to an HTML table using stargazer

Figures and Images in Markdown

Knitr will automatically include your images in markdown pages just as it did in LaTeX, but sometimes it's easier to manually include an image yourself.

```
![Alt text](/path/to/img.jpg)
```

```
![Alt text](/path/to/img.jpg "Optional title")
```

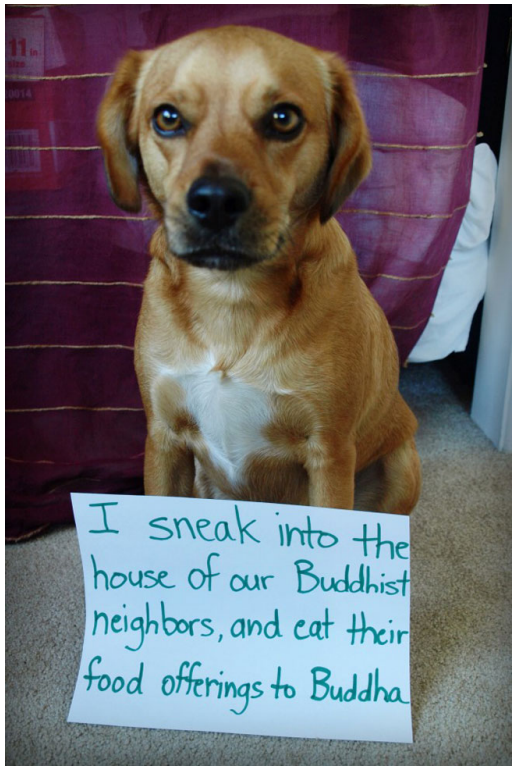
If you want to change the dimensions of the image, you may need to use HTML:

```
<img src="path/to/img.jpg" alt="some text"
      width="42" height="42">
```

Your Turn

Practice with images by inserting an image into the markdown document manually using both markdown syntax and HTML syntax.

If you don't have a favorite image, try this one



Presentations in Markdown using Rstudio

RStudio provides a feature to make HTML presentations in R as well using markdown.